

1. Given two integers n and m , find the multiple of m that is closest to n . If two multiples are equally close, return the larger one.

Test Case 1

Input: $n = 13$, $m = 4$

Output: 12

Test Case 2

Input: $n = 15$, $m = 6$

Output: 18

2. Given an array of positive integers and an integer K , find the length of the smallest contiguous subarray whose sum is at least K . Return 0 if no such subarray exists.

Test Case 1

Input: $arr = [2, 3, 1, 2, 4, 3]$, $K = 7$

Output: 2

Test Case 2

Input: $arr = [1, 1, 1, 1]$, $K = 5$

Output: 0

3. Given an array of integers, find the maximum product of any two numbers without sorting the array.

Test Case 1

Input: $[5, -10, -20, 3, 2]$

Output: 200

Test Case 2

Input: $[1, 7, 3, 4]$

Output: 28

4. Given a string, you may delete characters. Return the minimum number of deletions required so that no two characters have the same frequency.

Test Case 1

Input: "aaabbbcc"

Output: 2

Explanation: Frequencies are 3,3,2 $\rightarrow$ make them 3,2,1

Test Case 2

Input: "ceabaacb"

Output: 2

5. You are given a string containing only: '(', ')', '*'

The wildcard '*' can behave as: '(', ')'

or an empty string "". Return the length of the longest substring that can be made a valid parentheses sequence. A valid parentheses sequence means: Every opening bracket has a matching closing bracket. Order must be correct

Test Case 1

Input:

$s = "(*)"$

Output:

4

Explanation:

'*' can become '(' → string becomes "()" which is valid.

Test Case 2

Input:

s = "()*("

Output:

4

Explanation:

Longest valid substring is "()*"

'*' can become ')' → "()" valid length = 4

Test Case 3

Input:

s = "*)()(*"

Output:

4

Explanation:

Longest valid substring is "()" length = 4

Test Case 4

Input:

s = "(((*)"

Output:

4

Explanation:

Substring "(((*)" can be "()" valid length = 4